

Here are your notes on the Binomial Distribution, formatted for your VS Code Markdown editor.

Lecture Notes: The Binomial Distribution

1. What is the Binomial Distribution? 🤖

The Binomial Distribution is a **discrete probability distribution**.

It helps us answer questions like: "If I flip a coin 10 times, what's the probability I get *exactly* 7 heads?"

It models the number of **successes** (k) in a fixed number of **independent trials** (n), given that each trial has only two possible outcomes (a "success" or a "failure") and the probability of success (p) is the same for every trial.

Analogy: Orange vs. Grape Fanta 🧴

- **Question:** We survey 7 people. If there's a 50/50 preference ($p=0.5$), what's the probability *exactly* 4 people prefer Orange Fanta?
- **Trial (n):** Asking one person. We have $n = 7$ trials.
- **Success (k):** The person prefers Orange Fanta. We want $k = 4$ successes.
- **Probability (p):** The chance a random person picks Orange. We assume $p = 0.5$.
- **Failure:** The person prefers Grape Fanta. The probability is $(1 - p) = 0.5$.

2. Conditions for a Binomial Distribution

You can only use the Binomial model if your experiment meets four conditions. A good acronym is **BINS**:

- **B - Binary:** Each trial has only two possible outcomes (e.g., success/failure, heads/tails, orange/grape).
- **I - Independent:** The outcome of one trial does not affect the outcome of any other trial. (One person's Fanta choice doesn't change the next person's probability).
- **N - Number:** The number of trials (n) is fixed in advance. (We decided to ask 7 people).
- **S - Success:** The probability of success (p) is the same for every single trial.

3. The Formulas

There are two key parts:

1. Calculating the *number of ways* an event can happen.
2. Calculating the *probability* of any one of those ways.

Part 1: The Binomial Coefficient (Combinations)

This answers: "How many different ways can I get k successes in n trials?"

It's written as $\binom{n}{k}$ and read "n choose k".

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- $n!$ means "n-factorial" (e.g., $4! = 4 \times 3 \times 2 \times 1$).
- **Example (3 people, 2 prefer orange):** How many ways can 2 out of 3 people prefer orange?

- $n = 3, k = 2$
- **ParseError: KaTeX parse error: Expected group after '_' at position 24: ...}{2} = \frac{3!_{2!(3-2)!} = \dots**
- This matches our manual count: (Orange, Orange, Grape), (Orange, Grape, Orange), (Grape, Orange, Orange).

Part 2: The Binomial Probability Mass Function (PMF)

This formula gives the final probability of getting *exactly* k successes.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Let's break it down:

- $\binom{n}{k}$: The *number of ways* it can happen (from Part 1).
- p^k : The probability of getting k successes (e.g., 0.5^4 for 4 orange lovers).
- $(1 - p)^{n-k}$: The probability of getting $(n - k)$ failures (e.g., 0.5^3 for 3 grape lovers).

The formula simply multiplies the **(Number of Ways)** by the **(Probability of one specific way)**.

4. Full Example: 4 out of 7 Fanta Fans

Let's solve the video's problem: What's the probability 4 out of 7 people prefer orange, if $p = 0.5$?

- $n = 7$ (total people)
- $k = 4$ (successes)
- $p = 0.5$ (probability of success)
- $(1 - p) = 0.5$
- $(n - k) = 3$ (failures)

Step 1: Find the number of combinations $\binom{7}{4}$

$$\binom{7}{4} = \frac{7!}{4!(3!)} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(4 \times 3 \times 2 \times 1)(3 \times 2 \times 1)} = \frac{5040}{(24)(6)} = \frac{5040}{144} = 35$$

- There are **35 different ways** for 4 out of 7 people to prefer orange.

Step 2: Find the probability of one specific way

- This is $p^k (1 - p)^{n-k}$
- $(0.5)^4 \times (0.5)^3 = (0.0625) \times (0.125) = 0.0078125$

Step 3: Get the final probability

- $P(X = 4) = (\text{Combinations}) \times (\text{Probability of one way})$
- $P(X = 4) = 35 \times 0.0078125 = 0.2734375$

So, there is a **27.3% chance** that you would see a 4-3 split just by random chance.

5. The Binomial Test & p-values

The **Binomial Test** uses this distribution to get a **p-value**.

- **p-value:** The probability of getting your observed result (e.g., 4/7) *or something even more extreme*, assuming the "no preference" model ($p=0.5$) is true.

- In the video's two-sided test, they calculated the probability of all possibilities that were as rare or rarer than 4/7.
- $P(X \geq 4) + P(X \leq 3)$
- This ended up being $0.5 + 0.5 = 1.0$.
- **Conclusion:** A p-value of 1.0 is extremely high. It means the 4-3 split is *perfectly consistent* with a 50/50 preference. We have no reason to reject the idea that both Fanta flavors are equally loved.